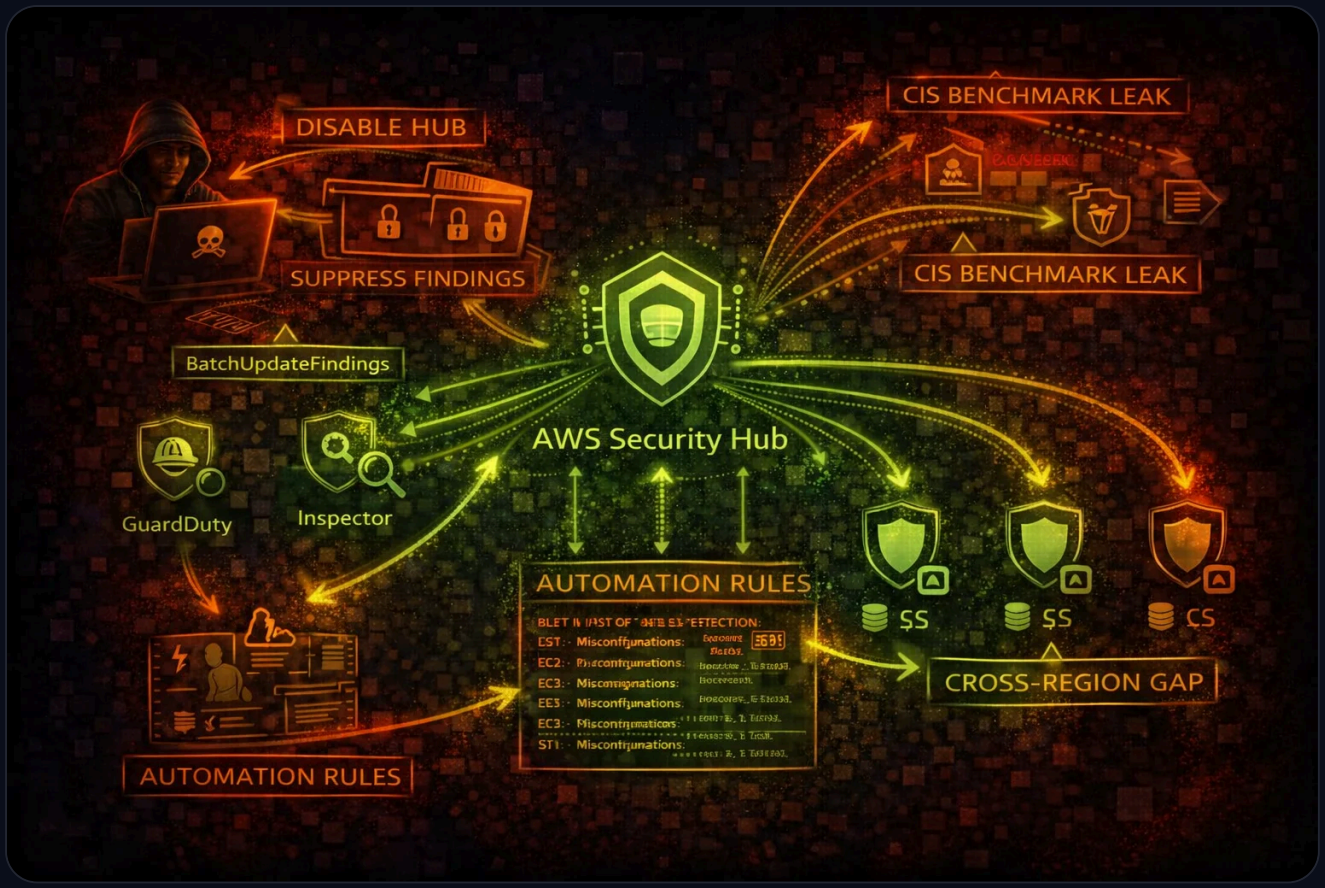


AWS Security Hub Security

SECURITY POSTURE

Security Hub aggregates findings from GuardDuty, Inspector, Macie, Config, and third-party tools. Attackers can disable it, suppress findings, or use GetFindings as a reconnaissance goldmine for misconfiguration discovery.

1 API	CIS/PCI	Multi-Svc	Recon
To Disable All	Compliance	Aggregation	Via Findings

Service Overview

Finding Aggregation

Security Hub ingests findings from GuardDuty, Inspector, Macie, Config, Firewall Manager, and third-party tools. Disabling Security Hub silences all integrated findings in a single API call, creating a massive blind spot.

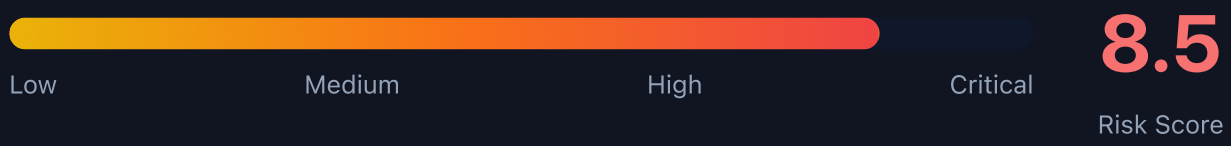
Compliance Standards

Built-in standards (CIS AWS Foundations, PCI DSS, NIST 800-53, AWS Foundational Security Best Practices) continuously evaluate your environment. GetFindings reveals every misconfiguration and compliance gap to an attacker.

Automation & Suppression

Automation rules can auto-update finding statuses. Attackers create rules to auto-suppress future findings, effectively creating persistent blind spots. BatchUpdateFindings can archive hundreds of findings instantly.

Security Risk Assessment



Security Hub is both a reconnaissance goldmine (GetFindings reveals all misconfigurations) and a key target for defense evasion (DisableSecurityHub, BatchUpdateFindings, CreateAutomationRule for suppression).

⚔️ Attack Vectors

Defense Evasion

- DisableSecurityHub silences ALL integrated findings
- BatchUpdateFindings suppresses findings covertly
- CreateAutomationRule auto-suppresses future findings
- DisassociateFromAdministratorAccount escapes org monitoring
- DisableImportFindingsForProduct blocks specific integrations

Reconnaissance via Findings

- GetFindings returns complete misconfiguration inventory
- CIS/PCI compliance failures map exploitable weaknesses
- Findings reveal public S3 buckets, open security groups
- Failed compliance checks identify unpatched resources
- Cross-region aggregation shows full org posture

⚠️ Misconfigurations

Coverage Gaps

- Security Hub not enabled in all regions
- Cross-region aggregation not configured
- Not all compliance standards enabled
- Third-party integrations not connected
- Findings not exported to external SIEM

Access Control Issues

- Member accounts can disable Security Hub independently
- No SCP preventing DisableSecurityHub
- BatchUpdateFindings allowed for non-admin users
- Automation rules not audited for suppression patterns
- Delegated admin account not hardened

🔍 Enumeration

Check If Security Hub Is Enabled

```
aws securityhub describe-hub
```

Get All Findings (Recon Goldmine)

```
aws securityhub get-findings \\
  --filters '{"ComplianceStatus":[{"Value":"FAILED","Comparison":"EQUALS"}]}' \\
  --max-items 100
```

List Member Accounts

```
aws securityhub list-members
```

List Enabled Standards

```
aws securityhub get-enabled-standards
```

List Automation Rules

```
aws securityhub list-automation-rules
```

🚨 Key Concepts

Finding Lifecycle Abuse

- NEW -> NOTIFIED -> RESOLVED -> SUPPRESSED workflow
- BatchUpdateFindings sets Workflow.Status to SUPPRESSED
- Suppressed findings hidden from default dashboard view
- RecordState set to ARCHIVED removes from active findings
- Compliance.Status override hides failed checks

Aggregation Architecture

- Region-scoped: each region runs independently
- Cross-region aggregation copies findings to one region
- Admin-member model: admin sees all member findings
- Disassociating from admin breaks centralized visibility
- Custom actions trigger EventBridge for automated response

⚡ Persistence Techniques

Persistent Suppression

- CreateAutomationRule to auto-suppress specific finding types
- Automation rule with broad criteria silences future detections
- Suppress by resource type to hide attacker infrastructure
- Modify existing automation rules to expand suppression scope
- Disable specific product integrations selectively

Monitoring Evasion

- Disable Security Hub in non-aggregation regions
- Disassociate member account from admin oversight
- Delete custom EventBridge rules for finding alerts
- Modify SNS subscriptions to redirect notifications
- BatchUpdateFindings to mark critical findings as resolved

🛡️ Detection

CloudTrail Events

- DisableSecurityHub - hub disabled entirely
- BatchUpdateFindings - findings suppressed/archived
- CreateAutomationRule - suppression rule created
- DisassociateFromAdministratorAccount - escaped monitoring
- DisableImportFindingsForProduct - integration disabled

Indicators of Tampering

- Security Hub disabled unexpectedly
- Large batch of findings suppressed at once
- New automation rules with broad suppression criteria
- Member account disassociated from admin
- Decrease in finding count without remediation

Exploitation Commands

Disable Security Hub

```
aws securityhub disable-security-hub
```

Suppress All Critical Findings

```
aws securityhub batch-update-findings \\
  --finding-identifiers '[{"Id":"arn:aws:securityhub:us-east-1:12345678901
2:finding/abc123","ProductArn":"arn:aws:securityhub:us-east-1::product/aws/
securityhub"}]' \\
  --workflow '{"Status":"SUPPRESSED"}'
```

Create Auto-Suppression Rule

```
aws securityhub create-automation-rule \\
  --rule-name "SuppressAll" \\
  --rule-order 1 \\
  --criteria '{"SeverityLabel":[{"Value":"CRITICAL","Comparison":"EQUAL
S"}]}' \\
  --actions '[{"Type":"FINDING_FIELDS_UPDATE","FindingFieldsUpdate":{"Workf
low":{"Status":"SUPPRESSED"}}}]'
```

Disassociate From Admin Account

```
aws securityhub disassociate-from-administrator-account
```

Recon: Get All Failed Compliance Checks

```
aws securityhub get-findings \\
  --filters '{"ComplianceStatus":[{"Value":"FAILED","Comparison":"EQUAL
S"}],"RecordState":[{"Value":"ACTIVE","Comparison":"EQUALS"}]}' \\
  --query 'Findings[].[Title,Resources[0].Id]' \\
  --output table
```

Disable Specific Product Integration

```
aws securityhub disable-import-findings-for-product \\
  --product-subscription-arn arn:aws:securityhub:us-east-1:123456789012:pro
duct-subscription/aws/guardduty
```

📜 Policy Examples

✕ Dangerous - Can Disable Security Hub

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": "securityhub:*",
    "Resource": "*"
  }]
}
```

Full Security Hub access allows disabling the service, suppressing findings, and creating suppression automation

✓ Secure - Read-Only Security Monitoring

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "securityhub:Get*",
      "securityhub:List*",
      "securityhub:Describe*"
    ],
    "Resource": "*"
  }]
}
```

Read-only access for security analysts to view findings without modification rights

✕ Dangerous - Can Suppress Findings

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": [
      "securityhub:BatchUpdateFindings",
      "securityhub:CreateAutomationRule",
      "securityhub:UpdateAutomationRule"
    ],
    "Resource": "*"
  }]
}
```

Ability to suppress findings and create automation rules enables covert defense evasion

✓ Secure - SCP Prevent Disable

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "PreventSecurityHubDisable",
    "Effect": "Deny",
    "Action": [
      "securityhub:DisableSecurityHub",
      "securityhub:DisassociateFromAdministratorAccount",
      "securityhub:DisableImportFindingsForProduct",
      "securityhub:DeleteMembers"
    ],
    "Resource": "*"
  }]
}
```

Organization SCP prevents member accounts from disabling Security Hub or escaping admin oversight

Defense Recommendations

SCP Prevent Disable

Use Organization SCPs to deny DisableSecurityHub, DisassociateFromAdministratorAccount in all member accounts.

Export Findings Externally

Send findings to S3 or external SIEM via EventBridge so they persist even if Security Hub is disabled.

```
aws events put-rule --name SecurityHubFindings \\
  --event-pattern '{"source":["aws.securityhub"],"detail-type":["S
ecurity Hub Findings - Imported"]}'
```

Enable in All Regions

Security Hub is regional. Enable in all regions and configure cross-region aggregation.

```
aws securityhub create-finding-aggregator \\
  --region us-east-1 \\
  --region-linking-mode ALL_REGIONS
```

Audit Automation Rules

Regularly review automation rules to ensure none are suppressing findings inappropriately.

```
aws securityhub list-automation-rules \\
  --query 'AutomationRulesMetadata[].{Name:RuleName,Status:RuleSta
tus}'
```

Use Org Delegated Admin

Centralize Security Hub management so member accounts cannot disable it independently.

```
aws securityhub enable-organization-admin-account \\
  --admin-account-id 123456789012
```

Alert on Security Hub Changes

Create EventBridge rules for DisableSecurityHub, BatchUpdateFindings, and CreateAutomationRule events.

Always obtain proper authorization before testing